

1. Create a structure to specify data on students given below:
Roll number, Name, Department, Course, Year of joining
Assume that there are not more than 450 students in the college.
(a) Write a function to print names of all students who joined in a particular year.
(b) Write a function to print the data of a student whose roll numbers are specified.
2. Create a structure to specify data of customers in a bank. The data to be stored is:
 - Account number
 - Name
 - Balance in account.Assume a maximum of 200 customers in the bank.

(a) Write a function to print the Account number and name of each customer with balance below Rs. 100.

(b) For withdrawal/deposit requests (fields: Acct. no, amount, code → 1 for deposit, 0 for withdrawal),
write a program to update balance. If on withdrawal the balance falls below Rs.100, display:
“The balance is insufficient for the specified withdrawal”.
3. An automobile company has serial number for engine parts starting from AA0 to FF9. The other characteristics of parts are year of manufacture, material and quantity manufactured.

(a) Define a structure to store this information.

(b) Write a program to retrieve information on parts with serial numbers between BB1 and CC6.
4. A record contains name of cricketer, his age, number of test matches that he has played and the average runs that he has scored in each test match. Create an array of structures to hold records of 20 such cricketers and then write a program to read and arrange records in ascending order by average runs.
5. Define a structure Employee containing:
 - Employee code
 - Name
 - Date of joiningWrite a program to enter data into an array of employees. Ask user to input current date, then display names of employees whose tenure is ≥ 3 years.

6. Create a structure Library to hold:

- Accession number
- Title of the book
- Author name
- Price
- Flag (issued or not)

Write a menu-driven program with options:

- a. Add book information
- b. Display book information
- c. List all books of a given author
- d. List title of a specified book
- e. Count books in the library
- f. List books in order of accession number
- g. Exit

7. Write a c program to Create an array of four function pointers where each pointer should point to a different function. Each function should take two integers and return a float. Using a loop, call function using the addresses present in the array.

8. Write a function that receives variable number of arguments, where arguments are coordinates of points. Based on number of arguments, display shape type (point, line, triangle, etc.).

9. Define a structure Date with members stored as bit fields:

- Day → 5 bits
- Month → 4 bits
- Year → 12 bits

Write a program to read date of joining of 10 employees and display them in ascending order of year.

10. Write a program to read and store information about insurance policy holder. The information contains details like gender, whether the holder is minor/major, policy name and duration of the policy. Make use of bit-fields to store this information.

11. Declare a structure Student containing:

- Name
- Age

- Address

Create and initialize 3 structure variables.

Define a function that converts all names to uppercase. Print updated records.

12. Write a C program to use an enum for traffic light states (RED, YELLOW, GREEN) and simulate a traffic signal system using switch-case and transition every N seconds.

13. Create an enum Permission { READ=1, WRITE=2, EXEC=4 };

Write a program to combine permissions using | (bitwise OR), and check permissions using &.

14. Define an enum for { ADD, SUB, MUL, DIV, EXIT } and Write a calculator program where menu selection maps to the enum and corresponding function.

15. Define a struct and a union with same members (int, float, char).

Write a program to print their sizes and explain difference in memory allocation.

16.

```
struct student { int r; float m; };

main() {

    struct student v;

    printf("%d\n", sizeof(v));

    printf("%d\n", sizeof(struct student));

}
```

17.

```
#include <stdio.h>

struct A { int x; int y; };

main() {

    struct A v = {10,20};

    struct A *p = &v;

    printf("%d %d\n", p->x, p->y);

}
```

18.

```
#include <stdio.h>

struct A { int x; int y; int *p; };

main() {

    struct A m= {10,20};

    m.p = &m.x;

    *(m.p) = 100;

    m.p = &m.y;

    *(m.p) = 200;

    printf("%d %d\n", m.x, m.y);

}
```

19.

```
#include <stdio.h>

struct A { int x = 10; int y = 20; } mchip;

main() {

    printf("%d %d\n", mchip.x, mchip.y);

}
```

20.

```
#include <stdio.h>

struct st { int a:5; char b:3; int c:2; };

main() {

    struct st mchip;

    printf("%d\n", sizeof(mchip));

    printf("%u\n", & mchip.a);

    printf("%u\n", & mchip.b);

    printf("%u\n", & mchip.c);

}
```

}